



Document Effective Date
04/06/2025

IS WASTE MANAGEMENT PROCEDURE

DOC NO: IS-HSE-PRC-012

Rev: 02

Quality Reviewed By: OHO/4(I)

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FINAL APPROVAL		
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1.0 PURPOSE

This waste management procedure provides a structured approach to identifying, classifying, handling, segregating, storing, transporting, and disposing of hazardous and non-hazardous waste generated in offshore operations. It aligns with QatarEnergy corporate standards, procedures, and guidelines while ensuring compliance with Qatari regulations and international agreements.

This waste management procedure identifies the appropriate waste disposal management practices for each type of waste generated in the Idd El-Shargi offshore operations and the RLIC Mud Plant.

This procedure provides a guideline to manage various type of waste i.e non- hazardous, hazardous and domestic. The guidelines are in line with:

- Relevant legislation of the State of Qatar Environmental Regulations.
- Relevant International Agreements ratified by the State of Qatar
- Consent to Operate conditions.
- Corporate Standard for Waste Management (CORP-ENV-STD-031), Corporate Procedure for Waste Management (QP-ENV-PRC-005)
- Corporate Guideline for Waste Management (QP-ENV-GDL-006)

2.0 SCOPE AND APPLICABILITY

This Procedure applies to all waste generated by QatarEnergy-IS and RLIC Mud plant. This is excluding operations at Halul Island where routinely generated wastes are managed by QatarEnergy separately.

This includes:

- Waste Classification
- Waste reporting through SAP/QatarEnergy Connect
- Methods of handling, segregation, storage, transport, treatment and disposal.

3.0 ROLES AND RESPONSIBILITIES

The roles and responsibilities are structured to ensure effective waste management, with defined duties for waste generator, Logistic & Materials Supervisor, Materials Clerk (Offshore)/ Waste Initiator and HSE Environment teams.

WASTE GENERATOR.

- Ensure compliance to general identification & classification of waste.
- Ensuring compliance to waste handling, storage, labelling prior to transportation
- Ensure adherence to relevant company standard, procedure and guideline for waste management.
- Ensure the requirements for additional documents are accomplished adequately to process the transfer/disposal of hazardous waste such as Naturally Occurring Radioactive Material (NORM) survey certificate by RPO and details of source of waste and potential contamination.

LOGISTIC & MATERIALS SUPERVISOR

- Maintain hazardous waste and special wastes storage, shipping and disposal documentation.
- Update the waste inventory register to indicate the date hazardous wastes are shipped for disposal.
- Assist the Environmental team in arranging, preparing and shipping hazardous waste to disposal locations.

MATERIALS CLERK (OFFSHORE)/ WASTE INITIATOR

- Prepare cargo manifests for Inert Non-Hazardous and Recyclable Metals skips.
- Prepare cargo manifests for hazardous waste skips using the Waste Tracking Register.
- Ensure that hazardous waste drums have an RLIC Hazardous Waste label completed and affixed to the drums prior to shipping.
- NORM Contaminated wastes shall be properly labelled, segregated and to be transported to Dukhan for temporary storage through RLIC.
- Ensure that NORM contaminated waste drums have a completed Hazardous Waste label, and an international trefoil radiation label affixed to the drums prior to shipping.
- To ensure the waste handling and segregation during storage.
- To identify classification of waste generated with the information from operation team.
- Use the Waste Reporting through QatarEnergy Connect to manage waste reporting including request of initiation, approvals, transport, and storage of waste.
- Ensure inspection of waste to avoid miss categorizing waste or potential recycling opportunity

ENVIRONMENT TEAM

- Provide guidance and advice regarding management and disposal of any waste streams that have not been identified in this plan.
- Review and update the IS-HSE-PRC-012 Waste Management Procedure every 3 years or as necessary.
- Contract disposal services for hazardous waste and special wastes.
- Submit required reports to MoECC and QatarEnergy regarding waste storage, shipping and disposal.
- Prepare waste related report as per CTO requirement.

4.0 PROCEDURAL STEPS

To apply this procedure, take the following steps:

1. Identify waste type and proper classification. Ensure all relevant information such as MSDS, materials of waste generated or potential contamination are properly captured.
* Refer to "Handling and Disposal Practices by Waste Stream" in Appendix B for guidance.
Select the best disposal practice from the options given. If guidance for the particular waste is not available, refer to the HSE Department. Ensure waste are being store and segregated according to QatarEnergy color-coded system.
2. Ensure appropriate labelling and documentation is available to accompany the waste transfer or as when requested for clarification.

3. Transport waste to designated disposal facilities in compliance with regulations.
4. If the guidance in this Waste Management Procedure is no longer considered valid, inform the HSE Department.
5. Ensure the material is correctly labelled, including a Ras Laffan Industrial City Hazardous Waste Identification Label (Appendix A) if being transported via Ras Laffan.
6. Use the cargo manifest if shipping the waste offsite.
7. Supply Chain shall maintain Hazardous Waste documentation denoting generation, quantity, shipment and final disposal of all hazardous and special wastes.

4.1 IDENTIFICATION AND CLASSIFICATION OF WASTE

Waste generator shall properly identify and classify the type of waste that going to be disposed. All type of waste to be classified in following categories. Refer to Appendix B for details classification.

4.1.1 Non- Hazardous Waste

Is defined as any solid, semi-solid or liquid material, which does not have any adverse impact on the environment or to human health in an uncontained or untreated state. These types of waste are segregated according to its characteristic and recycled whenever possible, otherwise these are considered as non-recyclable domestic wastes. Examples of these non-hazardous wastes generated in QatarEnergy Idd El Shargi Operations can be any or a combination of the following wastes:

- a. Non-Recyclable, Non-Hazardous Wastes
 - Domestic refuse
 - Inert / Industrial Waste
- b. Recyclable
 - Kitchen and Canteen Wastes
 - Scrap Wood / Wooden Crate / Wooded Boxes
 - Rubber Tires
 - Used Paper and Cardboards
 - Plastic
 - Metal and metallic wastes
 - Construction Wastes

4.1.2 Hazardous waste

Any hazardous waste shall be identified on the manifest with its primary hazard and secondary hazard listed. If a material is flammable and a poison, both hazards shall be in the description and underlined so that the warning stands out, e.g., Flammable, Poison.

The MoECC stipulates that classification of hazardous wastes in Qatar shall be according to the Basel Convention 1992, a treaty Qatar has ratified, characterized as follows:

- Explosive
- Flammable
- Oxidizing
- Infectious Substances
- Corrosive
- Toxic
- Ecotoxic

Use the following hazard descriptions:

Hazard	Description
Corrosive	Acids, bases, and anything that causes visible destruction or irreversible alterations in human skin tissue or that has a severe corrosion rate on steel.
Flammable	Flammable liquid as a liquid having a flash point of not more than 93 Celsius. i.e. hydrocarbon, solvent etc.
Oxidizer	Substances such as chlorate, permanganate, peroxide, or a nitrate, that yields oxygen readily to the environment.
Explosive	the property of a substance i.e. its ability to explode, release of energy, any substance or device that can be made to produce volume of rapidly expanding gas in an extremely brief period
Poison/ Toxic	Poisonous gases or liquids of such a nature that a very small amount of the gas, or vapour of the liquid mixed with air is dangerous to life. Examples would be hydrogen sulphide or hydrogen cyanide.
Flammable	Describes a liquid, gas or semi-solid or solid which has a flash point below 60°C (140°F).
Medical Waste/ Infectious Substances	Includes sharps (needles, syringes, and scalpels), bandages and any other material which is contaminated with blood or other bodily fluids. These wastes are collected separately at our clinic locations and placed in special containers designed for storage, handling and disposal of medical wastes.
Ecotoxic	Substance that may cause harm to environment i.e. marine, ozone and aquatic life.
NORM Contaminated Wastes and Materials	Waste (sludge, solids) and materials (equipment, tubulars, etc.) identified as NORM contaminated (based on requirements found in IS-PRC-PRC-247) shall be shipped as Hazardous Waste and labelled with a Trefoil radiation symbol.
Irritant	Liquid or solid which upon contact with fire or when exposed to air gives off dangerous or intensely irritating fumes, or can irritate skin and eyes through contact but is not listed as a poison.

For complete definition of the above, please refer to Corporate Standard (CORP-ENV-STD-031), Corporate Procedure for Waste Management (QP-ENV-PRC-005) and Corporate Guideline for Waste Management (QP-ENV-GDL-006).

Examples of these hazardous wastes generated in QatarEnergy Offshore Operations can be any of the following wastes:

- Empty Chemical and Oil Drums – may contain traces of hazardous chemicals or oil that are hazardous in nature.
- Empty Chemical Containers in pail (20-25L)
- Used Lead – Acid Batteries
- Used Nickel – Cadmium Batteries
- Used Acids (Hydrochloric, Sulfuric Acids)
- Expired Chemicals
- Oily Sludge - (NORM Free)
- Oil-contaminated Sand/ Debris – (NORM Free)
- Used Oil / Dirty Oil / Slop Oil

4.1.3 Radioactive Waste.

The Basel Convention excludes the management of radioactive wastes which are covered under the International Atomic Energy Agency. The following document shall be consulted when dealing with radioactive materials and radioactive wastes in Idd El Shargi Operations:

- Procedure for IS Handling Norm Contaminated Materials & Equipment IS-FOP-PRC-247.

4.2 WASTE HANDLING AND DISPOSAL PRACTICES – BY WASTE STREAM

Regulatory controls over waste disposal consist of State and International Law. Additionally, there are Environmental Protection Guidelines, issued by the Ras Laffan Industrial City management, which are applicable to all business operations within the city boundaries and which apply to QatarEnergy-IS by virtue of our Ras Laffan logistic.

QatarEnergy-IS shall inform RLIC of all storage, movement and eventual disposal of all waste material. Additionally, waste material and hazardous cargoes require manifesting in accordance with the RLIC guidelines and identification using RLIC waste labels.

Appendix C provides a flowchart summary of waste disposal options.

4.3 MANAGEMENT PRACTICE BY WASTE STREAM

Appendix B lists routine waste streams generated by QatarEnergy-IS operations. Each waste stream is described by name, composition, handling and disposal hazards, preferred PPE and preferred disposal method. This list shall be a reference and starting point for the proper management and disposal of wastes. Contact the HSE Department if assistance is required.

4.4 MOVEMENT OF WASTE WITHIN QATAR

The QatarEnergy-IS cargo manifest shall be used to track waste. A separate cargo manifest (see example in Appendices 4 and 5) shall be completed for waste shipments to allow tracking of the waste. Hazardous waste shipping manifests shall be generated. This enables QatarEnergy-IS to document and track all hazardous wastes generated offshore which are shipped ashore to the RLIC logistic temporary/transition hazardous waste storage area.

The Logistic Supervisor maintains records of waste manifests of waste received from offshore and shipping manifests for waste sent for disposal. This is essential for general control of waste disposal and also to allow accurate reporting, when necessary, to the regulatory bodies (e.g., the MOECC requires quarterly reporting of waste inventories).

4.5 EXPORT OF WASTE

Wastes generated by QatarEnergy-IS for which no waste management option currently exists in Qatar shall be stored pending development of an appropriate management solution within Qatar or will be exported for treatment or disposal in another country.

In the case of wastes which are exported, QatarEnergy-IS shall comply with all provisions of the Basel Convention on the Control of Trans Boundary Movements of Hazardous Wastes and Their Disposal (as amended). This Convention, inter alia, specifies a particular manifest system for transboundary shipment of waste.

4.6 SHIPPING SKIPS - QATARENERGY-IS COLOUR CODE

Type	Colour	Description
Inert Non- Hazardous Waste	Orange	Office waste and general garbage, wood, building debris, rubber, cotton, dust filters, cable insulation, plastics
Recyclable Metals	Red	Pipes, bolts, valves, aluminium cans, electrical parts and cables, gauges, motor bearings, grinding wheels, slings and shackles
Hazardous Waste	Black	Full/partially full drums of chemicals, hydrocarbon sludge/waste, aerosol cans, filter elements, used/empty chemical containers, empty paint cans, used batteries, used oil gaskets, non-asbestos insulation materials

The labels shown in Appendix D shall be affixed to all waste skips.

4.7 WASTE REPORTING/ LOGIN VIA QATARENERGY CONNECT

- a. Waste Initiator shall use the Waste Reporting through QatarEnergy Connect which is a computer module developed to manage waste reporting including request of initiation, approvals, transport and storage of waste from one location to another or within the same location.
- b. Waste Reporting on QatarEnergy Connect for Offshore and Halul is outlined/ summarized in APPENDIX-C.
- c. Waste Initiator and controller shall consider the following important reminders before wastes are reported and processed in QatarEnergy Connect:
 - Prepare a separate Waste Consignment Notes for:
 - Non-Hazardous Wastes
 - Hazardous Wastes
 - Scrap Metal
 - Waste Oil
- d. Do not mix the wastes that are heading to different location.

4.8 HAZARDOUS WASTES GOING TO MIC-HWTC

For the hazardous wastes going to MIC-HWTC following are required:

- A filled-up Waste Profile Sheet shall be submitted to MIC. The Waste Generator and Environment Team shall arrange to prepare the required form to be submitted to MIC-HWTC. Please refer to Appendix E for a sample copy of this form.
- At least one kilogram of sample shall be sent along with the waste to facilitate the analysis of waste once it reached RLIC.
- Material Safety Data Sheets (for expired chemicals)
- Photos (if necessary) of the wastes to be transported.

- Hazardous waste that is being stored in PS1 shall be classified and sample shall be send to MIC-HWTC to confirm their acceptance.
- If the hazardous waste is not accepted by MIC HWTC, environment team shall coordinate the disposal through available MoECC approved waste contractor.

4.9 REQUESTING HAZARDOUS AND SPECIAL WASTE DISPOSAL FROM MOECC

Occasionally it will be necessary to dispose of waste which does not fall into the MoECC non-- hazardous and industrial waste categories. The process shall be coordinated through the HSE Department. An MSDS or a lab analysis showing the composition of the waste shall be sent with the form to the MOECC for approval.

4.10 TRAINING

Training is provided in accordance with the QatarEnergy-IS Training Matrix. All personnel shall receive general waste management training. Additional training is provided to select offshore production staff and onshore logistic staff whose job duties require more in-depth knowledge of elements of this procedure

4.11 SAMPLING AND ANALYSIS

It is sometimes necessary to conduct sampling and analysis of waste streams in order to receive approval from the MOECC or the disposal facility. When this is necessary samples shall be obtained and sent to qualified laboratories to conduct the required analytical procedure.

4.12 STORAGE

QatarEnergy-IS is required by MOECC regulations and Consent To Operate (CTO) to safely store all hazardous solid waste in a contained facility onsite for a maximum period of 90 days, until such wastes can be treated to a non-hazardous state or exported to a treatment facility.

Total quantity of solid waste generated on a Quarterly / Yearly basis (in metric tonnes) shall be forwarded to the MOECC with the Quarterly / Yearly Environmental Monitoring report. This report shall have a breakdown of solid waste with three major categories viz., Hazardous, Non-Hazardous and Inert.

All unidentified chemicals shall be treated as hazardous waste and stored accordingly i.e., in the Hazardous Waste Storage Area, until their composition and disposal mechanism have been determined.

4.13 WASTE DISPOSAL CONTRACTORS

QatarEnergy-IS generated wastes shall only be transferred to waste disposal contractors which are approved by HSE, and then only if those waste management contractors hold all relevant permits and providing those permits are 'current'.

Exceptional cases such as NORM or other non-treatable waste can be exempt by agreement with MoECC.

All unidentified chemicals (except environmental friendly chemicals) shall be treated as hazardous waste and stored accordingly i.e., in the temporary Waste Storage Area in PS-1, until their composition and disposal mechanism have been determined.

- Inspection of waste by Environment team will be done on necessity & request basis.

4.14 IDENTIFICATION OF NEW WASTE STREAMS

If a waste is identified that is not included in this Waste Management Procedure, it shall be categorized by determining the properties of the waste. The HSE Department shall be responsible for waste categorisation.

Correct categorisation of a waste stream is important to reduce waste management costs and potential for liability.

Qualitative or quantitative measures may be employed to determine the waste category when:

- There is some doubt over the appropriate classification.
- When the source and composition of the waste is not known.
- When it is uncertain whether the waste needs to be segregated from other waste types.

In some instances, it will be necessary to conduct laboratory testing to determine the waste characteristics and classification (see 4.8 Sampling and Analysis).

5.0 REFERENCES TO OTHER DOCUMENTS

Title	Document Number	Internal/ External Doc	Document Referencing
Handling NORM Contaminated Materials and Equipment	IS-FOP-PRC-247	Internal	Sideward
Procedure for working with Pyrophoric Scale	IS-FOP-PRC-268	Internal	Sideward
HSE Training Procedure	IS-HSE-PRC-002	Internal	Sideward
Corporate Standard for Waste Management	CORP-ENV-STD-031	Internal	Upward
Corporate Procedure for Waste Management	QP-ENV-PRC-005	Internal	Upward
Corporate Guideline for Waste Management	QP-ENV-GDL-006	Internal	Upward
Procedure for Control of Asbestos	IS-HSE-PRC- 040	Internal	Sideward

6.0 DEFINITIONS

Term	Definition
Waste	Waste is any solid, liquid or gaseous material, which is discharged to the environment or to the plant boundary not as a product for sale. It is material that will require further treatment or in terms of a material balance would show up as a loss to the process. Waste is any unwanted substance or object, which is disposed of, is intended to be disposed of, or is required to be disposed of by the provisions of national law.
Inert Non-hazardous Waste	Solid waste that has no active chemical or biological properties. These wastes do not undergo environmentally significant physical, chemical or biological transformation and have negligible potential to cause environmental harm.
Hazardous Waste	Any waste that has the potential to cause harm to personnel, the environment, or assets.
Special Waste	Waste which does not fit into the Hazardous, Inert Non- hazardous, waste categories or which has special waste disposal practices. Naturally Occurring Radioactive Material (NORM) waste and asbestos waste fall into this category.

Term	Definition
Low Toxicity Oil and Gas Waste	Describes most of the high-volume low toxicity wastes generated in oil and gas drilling and production and includes produced water, drill cuttings, drilling fluids and all other materials from the wellbore associated with the production of oil and gas.
Pollution Prevention	The reduction in volume and/or toxicity of waste prior to discharge or disposal. Prevention primarily includes source reduction and recycling.

APPENDIX

APPENDIX A-HAZARDOUS WASTE IDENTIFICATION LABEL FOR RLIC

WASTE	
Qatar Law Prohibits the Improper Disposal	
Waste Name: _____	
Category:	HAZ ☐ Non-HAZ ☐
MSDS Provided: YES ☐ NO ☐ Physical State: Solid / Liquid / Sludge	
Hazards :Flammable ☐ Oxidizer ☐ Corrosive ☐ Toxic ☐ Others ☐	
If, Other Please Specify _____	
Drum No. _____ of _____	Date: _____
Generator's Name: _____	
Waste Generator's Identification No.: _____	
Generator's Contact No.: _____	
Handle With Care Spillage May Cause Ground Water / Soil Contamination	

APPENDIX B - WASTE STREAM INFORMATION AND MANAGEMENT

1	Abrasive Blasting Media (Grit)
Description	An accumulation of used blasting grit, paint particles and trace amounts of abraded metals.
Sources	Metal surface preparation for painting / fabrication / maintenance
Category	Normally a non-hazardous waste but shall be considered hazardous if the lead (Pb) content is above 5mg/l (milligrams/kg).
Handling and Disposal	Wear respiratory protection when handling. If elevated lead (Pb) levels (above 5 mg/kg) are encountered, then treat as a hazardous waste. Drum or bag the used grit and paint chips and transport to shore with containers labelled "USED GRIT & PAINT CHIPS - May contain lead (Pb)".
Preferred PPE	If Lead -This equipment would include but not be limited to: Disposable or cleanable work gloves Coveralls (Tyvek or similar) with foot covering Goggles or face shields Properly fitted half-face respirators with HEPA cartridges If NO Lead- Cotton Gloves, Safety Glasses and dust mask
Preferred Disposal Method	Basel Waste Management Code: D1 Disposal of blast media used by contractor remains the responsibility of that contractor. Collect and ship ashore for disposal by landfill in an approved non- hazardous waste disposal site (hazardous waste disposal site if containing Lead).
Grit Waste Management	Ministry Of Environment - Consent to Operate Document S.C.E 0017/ 557/ 07 - Requires that during grit blasting operation 70% of the grit used, is captured.

2	Acid (Spent)
Description	Refers to acidic solutions that have been used to chemically treat a wellbore or a particular geologic strata or formation. Normally, this material will have been pumped into the wellbore and returned to the surface.
Sources	Treatment of wellbore.
Category	Hazardous unless pH > 2

2	Acid (Spent)
Handling and Disposal	Handle in a manner which prevents spillage and contact with people. Storage at source in corrosion proof container such as plastic drum/barrel - label "Acidic Waste - Corrosive", label as "Hazardous" if pH not > 2.
Preferred PPE	Neoprene Gloves, Goggles or Face Shield, Chemical Apron, Half Mask Cartridge Respirator
Preferred Disposal Method	Basel Waste Management Code: D9 + D3 Treat (neutralize) so pH is in the level 6-9 and send with produced fluid to Halul Island. Re-inject into a disposal or injection well.

3	Antifreeze (Used)
Description	Used antifreeze arises from both standard automobile/truck engines and certain offshore process engines. The waste can contain elevated levels of various metals as a result of high temperature contact with the engine.
Sources	Vehicle / process engine maintenance.
Category	Hazardous waste
Handling and Disposal	Handle in a manner which prevents spillage and contact with people. Storage at source in corrosion proof container such as plastic drum/barrel - label "Glycol Waste - Poison".
Preferred PPE	PVC, Latex, Nitrile or Neoprene Gloves, Goggles or Face Shield, Particle Mask
Preferred Disposal Method	Basel Waste Management Code: D9 / D10 For vehicles: it is the responsibility of the leasing company to handle fluid changes. For QatarEnergy-IS owned vehicles use a local garage. For offshore: used antifreeze shall be injected into the produced water stream and hence to Halul Island. The other preferred option is to drum the fluid and ship it ashore.

4	Asbestos containing material
	See procedure: IS-HSE-PRC-040 Control of Asbestos
5	Batteries - Lead Acid {Used}
Description	Lead acid (accumulator) batteries are used in vehicles and static plant such as generators.
Sources	Vehicle and static plant maintenance activities.
Category	Hazardous waste due to acid content and lead sludge content.
Handling and Disposal	Used batteries shall be stored in well-ventilated area. Acid and residual sludge shall be drained (and flushed with water) from batteries prior to shipping. Battery cases may then be regarded as non-hazardous. Acid / sludge then is regarded as a "Spent Acid Waste" (3.2.2) containing lead and treated accordingly
Preferred PPE	Neoprene Apron, Neoprene Gloves, safety glasses or goggles AND a face shield
Preferred Disposal Method	See 3.2.2 for acid sludge treatment. Basel Waste Management Code: R4 Battery cases to be returned to vendor where practicable.

5b	Dry Cell / Ni-Cd Batteries
Description	Dry cell / Ni-Cd batteries originate from many different sources including flashlights, meters, radios, emergency lighting systems, instrumentation etc. Some batteries are recyclable.
Sources	Consumable items in electronic/ electrical equipment.
Category	Non-hazardous waste unless containing mercury.
Handling and Disposal	9-volt batteries shall be fully discharged as a short across the terminals could lead to the generation of heat and thereby result in a fire in a general waste skip.
Preferred PPE	If Leaking, then safety glasses and neoprene gloves. If not leaking, no PPE is necessary
Preferred Disposal Method	Basel Waste Management Code: R4 (preferred) / D1 Small quantities may be disposed of with refuse by placing in normal trash; this includes batteries used in a flashlight or portable radio, e.g. A, AA, AAA, and 9 volts (fully discharged)

6	Drill Cuttings
Description	Drill cuttings are the formation rock and sediment brought to the surface by circulating drilling mud. If the cuttings are oil contaminated in excess of the levels permitted by ROPME or any Consent to Operate, then they shall be regarded as oil contaminated waste as described in Section 3.2.16
Sources	Drilling of wellbores.
Category	Low toxicity oil and gas waste, non-hazardous waste unless significant oil content.
Handling and Disposal	May be discharged directly to the sea so long as it does not produce a sheen.
Preferred PPE	Safety glasses and rubber gloves, Apron
Preferred Disposal Method	Basel Waste Management Code: D7 provided insignificant oil content, otherwise D9+D3

7	Drilling Fluids
Description	Refers to the various drilling mud systems and their associated additives: weighting agents, surfactants, muds, acids, inhibitors, gels, solvents, and other materials.
Sources	Drilling of wellbores.
Category	Low toxicity oil and gas waste, non-hazardous waste unless significant oil content.
Handling and Disposal	May be discharged directly to the sea so long as a sheen is not produced. The pH shall be between 5 and 10. "The heavy metal content of barite (dry powder (mg/kg) shall be monitored every 6 months and the results recorded. Discharge shall be within the General Industry Standards". Cadmium < 3 mg/kg in barite (dry powder) Mercury < 1 mg/kg in barite (dry powder)
Preferred PPE	Safety glasses and rubber gloves, Apron
Preferred Disposal Method	Basel Waste Management Code: D7 provided insignificant oil content, otherwise D9+D3

8	Hydrostatic Test Fluids
Description	Treated water utilized to pressure test a pipeline or vessel
Sources	Pipeline and vessel pressure testing (fabrication and maintenance).
Category	Non-hazardous waste except when containing certain additives, e.g. biocides
Handling and Disposal	Store in bulk storage tanks or rigid walled containers.
Preferred PPE	If Fluid Contains Biocide – Clothing: Gloves, coveralls, apron, and boots as necessary to prevent contact. Eyes: Chemical goggles: also wear a face shield if splashing hazard exists. Respiration: If vapours or mists are generated, wear a NIOSH/MSHA approved organic vapour/mist respirator or an air-supplied respirator as appropriate.
Preferred Disposal Method	Fresh water or sea water which is used as hydrostatic test fluid may be discharged to the sea if it meets the Regional Organization for the Protection of the Marine Environment (ROPME) requirements or if it has a variance issued from MOECC. ROPME prohibits discharge into the sea of fluids with oil and grease levels above 15 mg/l. Injection into a disposal well. Return to drums and ship ashore for disposal

9	Empty Drums
Description	Empty metal or plastic drums which previously contained chemicals or material used at a facility shall only be considered empty if all the contents have been removed using commonly employed practices. Follow the procedures below for handling all partially empty and empty drums.
Sources	Drilling chemical (and other chemical / material) supply and usage.
Category	Non-hazardous waste - if EMPTY and CLEAN If containing residues of hazardous chemicals or hazardous materials then containers shall be classified as hazardous waste.

9	Empty Drums
Handling and Disposal	Rough handling of drums, either when containing chemicals or when empty, shall be avoided. A label shall be present to clearly identify the previous contents. Bungs shall remain with drum before, during and after use. Empty the drum, to be sure all free liquid is removed. Refer to the Materials Safety Data Sheet (MSDS) for safe handling of the chemical. All chemical from the drum shall be collected and used, not drained as waste onto the ground or into the sea. Do not mix the remaining contents with drums containing different chemicals. Storage: Drums shall be stored in a location which will keep the drum on a rack or foundation preventing it from sitting in water and protected from rainwater. Drums shall, preferably, be stacked shall be on their side and shall be tightly bunged. The area beneath the drum storage shall have drip pans, bund walls or some other way of capturing drips, leaks or spills. Disposal: Empty drum as described above and Supply Chain department to send to Recycling Vendor or: Recondition/recycle through separate approved contractor or, Triple rinse, crush and place in metals bin for metal recycle or, Triple rinse. crush and take to approved landfill or, Drain, triple rinse, remove the label, cut out top and use as a trash container. Note: Where possible use the rinse water for the chemical's originally intended purpose. If not possible then the rinse water shall be treated as if it were waste product.
Preferred PPE	Safety glasses, leather or other cut resistant gloves. If drum contains residual chemical then use neoprene gloves, safety goggles, and apron.
Preferred Disposal Method	Return drum to vendor. Basel Waste Management Code: R4 / D1

10	Fluorescent Tubes and Compact Fluorescent Light Bulbs
Description	Used fluorescent tubes and fluorescent light bulbs require safe disposal. Breakage during storage and transport poses a hazard to persons and a risk of release of mercury to the environment.
Sources	Electrical lighting fixtures.
Category	Non-hazardous waste (under Qatari regulations).
Handling and Disposal	Used fluorescent tubes and light bulbs shall be returned to their original packaging or placed in a dedicated suitable container to reduce the likelihood of breakage during storage and transportation to onshore for disposal.
Preferred PPE	Safety glasses and cotton gloves
Preferred Disposal Method	There are no recycling facilities for used fluorescent tubes and fluorescent light bulbs in Qatar so they are placed in a non-hazardous land fill.

11	Iron Sulphide Scale (FeS)
	Please refer to PRC-268 Working with Pyrophoric Scale

12	Laboratory Chemicals
Description	Chemicals which are used in the offshore laboratory to perform laboratory tests. This includes samples which are analysed and the resulting mixtures of chemicals and samples.
Sources	Laboratory testing.
Category	May be hazardous or non-hazardous waste according to the properties of the chemical in question.
Handling and Disposal	Neutralize or treat and place in closed drain system. Do not discharge oily mixtures to sea.
Preferred PPE	As per the MSDS for each chemical used in the Lab. Minimum use of safety glasses required.
Preferred Disposal Method	Treatment necessary according to chemical composition. Seek HSE advice as necessary.

13	Medical Waste
Description	Typically includes used gauze, bandages, injection needles, medicine bottles and any material contaminated with blood or bodily fluids. Most of this waste is generated in the clinic offshore.
Sources	First aid activities and routine health care.
Category	Hazardous waste (unless known for certain to be non-infectious and non- hazardous).
Handling and Disposal	Treat all waste contaminated with bodily fluids as if it has blood borne pathogens ("universal precautions"). Place all needles, syringes, glass or anything sharp which could puncture the skin in a "sharps container"; a red or yellow, puncture resistant container, which has the biological hazard symbol on the side. Soiled bandages or dressings shall be placed in plastic bags, which also bear the biological hazard symbol and shall be labelled or tagged "Medical Waste" (Single, small adhesive dressings can be disposed of in the normal trash.) Medical waste is sent to shore, clearly identified as "Medical Waste" and segregated from regular waste. A chain of custody document tracks the movement of the medical waste from offshore to onshore. A waste disposal manifest tracks the waste to the disposal location.
Preferred PPE	If spilled, use disposable gloves, impervious disposable gown, face shield.
Preferred Disposal Method	High temperature incineration. Basel Waste Management Code D10

14	Mercury
Description	Mercury is a silvery, heavy liquid which can be highly toxic to humans and animals particularly as the vapour.
Sources	None known
Category	Hazardous waste
Handling and Disposal	Immediately contact the HSE Department if mercury is found. Do not handle without the proper personal protective equipment. Elemental mercury may be delivered to a reclaimer or given to a university or laboratory for re-use. Mercury vapour lights do emit mercury vapour when broken. Small amounts (1or 2 at a time) of mercury vapour lights can be disposed of by placing in regular trash skips. If substantial number of mercury vapour lights are to be disposed of at once then they shall be separated and shipped separately. Secure permission from the MOECC for disposal in a landfill. Note: It is not anticipated to find any elemental mercury at QatarEnergy-IS
Preferred PPE	Dimethyl mercury. This chemical is so toxic that its use always requires an SP. In addition, inner (silver shield, 4H, or other brand laminate-style gloves) and outer gloves (heavy-duty nitrile or neoprene, with long cuffs) shall be considered as a minimum when handling, using, or working with dimethyl mercury. For general mercury handling avoid skin and eye contact. Use rubber or plastic gloves when handling metallic mercury. DO NOT use lightweight disposable gloves for heavier jobs, as these can tear easily and allow mercury to lodge under the fingernails or contact other parts of the skin. When the operation may result in exposure to the face, wear splash-proof goggles and a face shield unless full-face respiratory protection is being used.
Preferred Disposal Method	Basel Waste Management Code: R4 Recovery of mercury

15	Metal Waste (Scrap Metal)
Description	Metal waste is the general term for basic metal and steel scrap. If the metal has scale, it shall be tested for NORM and Iron Sulphide. Examples of metal waste: old tanks, vessels, metal thread protectors, scrap pipe, casing, wireline metal beams, scrap valves, wiring, conduit, scrap rods, junk nipples, junk compressor parts, downhole equip., packers, drill Bits, sheet metal etc.
Sources	Construction, fabrication and maintenance activities.
Category	Non-hazardous waste unless containing NORM / Iron Sulphide.
Handling and Disposal	Segregate at source and place metal waste in appropriate red recycle skips to ship back to shore.
Preferred PPE	Safety Glasses, Heavy Leather Gloves, Steel Toed boots, Long Sleeve Coveralls
Preferred Disposal Method	Basel Waste Management Code: R4 Recycle with vendor (drill bits, packers) or with a metal reclaimer. If recycling is not practicable then landfill.

16	Methanol
Description	Methanol is a water-soluble chemical used as an antifreeze in maintenance activities. Additionally, it is used in wells and flow lines to prevent the formation of hydrates.
Sources	Maintenance.
Category	Normally a hazardous waste. (If used to hydrostatically test a production line or added to natural gas to prevent hydrates and it co-mingles with the product, it is technically not a waste.)
Handling and Disposal	When using methanol, waste shall be minimized by using all the methanol in a drum or container for its intended purpose. Sturdy drums and bulk containers shall be used in storing and handling methanol. All containers shall be labelled indicating contents. Notify the appropriate Supervisor immediately upon discovery of a spill. Also notify the HSE Advisor (HSE Department). If surplus methanol requires disposal, contact HSE Department.
Preferred PPE	Neoprene Gloves, Safety Goggles, Apron
Preferred Disposal Method	Basel Waste Management Code: D10 High temperature incineration.

17	NORM Contaminated Wastes
Description	Naturally Occurring Radioactive Material (NORM) wastes originate in the oil wells. Mineral solids contain traces of radioactive elements and their cousins. In QatarEnergy-IS wells the element most prevalent is Radium 226.
Sources	These solids mainly collect in equipment vessels such as Electric Submersible Pumps (ESPs) and separators. Some may form as scale in our piping and tubulars as well. The waste is generated from equipment clean out and replacement.
Category	Hazardous and/or Special Waste
Handling and Disposal	See procedure - PRC-247 - "Procedure for handling NORM contaminated materials and equipment. NORM contaminated sludge and solids shall be drummed in containers rated for liquids and sludges. Avoid allowing material to dry out and create dust. The radioactive solids, when dry, can be ingested by breathing. Observe PPE requirements and proper hygiene international radiation trefoil symbol label.
Preferred PPE	Disposable coveralls (Tyvek}, cotton gloves, P100 rated respirator, disposable boot covers
Preferred Disposal Method	Basel Waste Management Code: D3 or D10 combined with D12 Deep Well disposal or High temperature incineration with encapsulation and secure landfilling of radioactive residue and ash.

18	Oil Soaked Debris and Refuse
Description	Oil-soaked debris and refuse and inert solid waste which has become contaminated with crude oil is waste Examples of refuse and debris which may become oil soaked: paper, soil, fab shop rubbish, rags, damaged pigs, drill cuttings and well debris from rock catchers.
Sources	Normal operational activities, cleaning of spillages, maintenance activities.
Category	May be hazardous
Handling and Disposal	Segregate oil-soaked trash from non-oily trash. Allow oily trash to drain and/or weather as much as possible. Drain to the drain system and recover any oil.

	All storage containers shall be leak-proof. Containers for oily rags shall be closed to prevent spontaneous combustion. Use a trash skip for the drained trash. Recycle the recovered oil.
Preferred PPE	Safety Glasses, Neoprene gloves, use chemical apron if there is a large quantity of oil-soaked debris
Preferred Disposal Method	Basel Waste Management Code: D10 (D1 if no mobile hazardous constituents) High temperature incineration If low levels of oil contamination and analysis has proved removal of mobile constituents, then these wastes can be disposed of to an MOECC approved landfill

19	PCBs
Description	Polychlorinated Biphenyls (PCBs) were used as dielectric fluid in older transformers, capacitors and other electrical equipment. PCBs are a fire retardant. As PCBs are highly toxic, their use is being phased out.
Sources	Equipment maintenance/ de-commissioning.
Category	Special waste I Hazardous Waste. PCB-containing material is defined as any substance, mixture, or item with a concentration of 50 ppm PCBs or greater. Notify the facility supervisor and then the HSE Department immediately if PCBs are discovered.
Handling and Disposal	Operations All PCB equipment shall be labelled "PCB Equipment". Keep PCB equipment in service until ready to ship for disposal, if possible. Notify HSE immediately if any PCB equipment is taken out of service. Spill Response If PCB equipment is involved in a spill or incident of any kind, notify the facility supervisor immediately and then the HSE Department.

19	PCBs
Preferred PPE	Personal Protective Equipment (PPE) Personnel engaged in handling PCB-contaminated or PCB-impregnated materials (such as opening and restoring a felt containing ventilation duct joint or replacing the felt material within the joint, during which skin contact with PCBs is considered probable, shall wear the following: Nitrile or Viton® gloves Face shields vented chemical type goggles or other appropriate eye protective equipment wherever the possibility of eye contact exists. Dust removal operations within ventilation systems, either known or suspected of containing PCB-impregnated gasket material, require the use of the following personal protective clothing. Gloves. Latex gloves inside, covered by nitrile or Viton® gloves. If coveralls are worn, gloves shall be taped to coveralls. If conducting a visual inspection in which joints will be broken, but no cleaning will be accomplished, the nitrile or Viton® gloves shall be used. Safety goggles or face shield Disposable coveralls. Tyvek® with attached booties and hood. Coveralls need only be worn if the duct is physically large enough for bodily entry and will be entered. After ventilation cleaning work in which full personal protective clothing (coveralls) are worn, the following procedures shall be used to remove such clothing: Wipe any dust or debris remaining on the PPE using a wet rag. Place the rag in a plastic bag for disposal. Remove the tape and outer gloves and place in a plastic bag for disposal. Unzip coveralls and step out by turning the coveralls inside out. Place the coveralls in a plastic bag for disposal. Remove goggles and face shield. Clean goggles and face-shield. Remove the latex gloves and place in a Plastic bag for disposal.
Preferred Disposal Method	Basel Waste Management Code: D10 (with gas cleaning) PCBs shall be stored until suitable PCB disposal facilities exist in Qatar or exported for high temperature incineration or thermo-chemical treatment. Note: QatarEnergy-IS baseline safety and environmental sampling did not reveal any PCBs at the IS facility. In addition, the State of Qatar (MOECC) has banned any PCB imports.

20	Pigging Waste
Description	Waste generated from the pigging of oil and gas pipelines may consist of water, sludge, hydrocarbons, scale, iron sulphide and damaged pigs.
Sources	Pigging activities.
Category	Hazardous waste
Handling Disposal	Prevent spills or leaks. Capture in drip pans. Place hydrocarbons back into the production system. Place rest of liquids into open drain/ sump system. If iron sulphide is suspected then keep wet and refer to the iron sulphide section 3.2.10.
Preferred PPE	If Hazardous, then use SCBA, Neoprene Gloves, Chemical Apron
Preferred Disposal Method	Basel Waste Management Code: D9 (separation) / R3 / D10 Where practicable, return hydrocarbon liquids to production system. Collect solids and ship ashore in plastic drums with absorbent material for disposal by incineration or landfill depending on content. Place aqueous liquids in drain system.

21	Produced Water
Description	Produced water is produced with and separated from the oil and/or gas. It may contain other constituents, either naturally occurring (e.g., traces of hydrocarbon, H ₂ S) or introduced (e.g., well treatment fluids). Produced water can be re-injected into a production zone, a non-producing zone, or disposed of overboard. If it doesn't exceed 40 mg/l (monthly average) and a 100 mg/l for any excursion.
Sources	Oil and gas production.
Category	Low toxicity oil & gas water waste.
Handling and Disposal	All produced water tanks and piping systems shall be free of leaks and protected from external corrosion. Produced water leaks shall be controlled and contained at the time of discovery, just as oil spills. Spill prevention plans shall be referred to for guidance.
Preferred PPE	If potentially contains H ₂ S then use SCBA or airline respirator, Neoprene Gloves and Coveralls
Preferred Disposal Method	Basel Waste Management Code: D7 / D9+D10 Re-injection into a producing or disposal zone. Shipped with oil to Halul Island.

	Alternatively, if the above is not practicable the waste may be injected into company-controlled disposal well.
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22	Pesticide, Herbicide and Biocide Wastes
Description	The liquid or solid chemicals left over after the use of pesticides, herbicides or biocides shall be treated as hazardous waste.
Sources	Domestic
Category	Hazardous waste
Handling and Disposal	Single cans, when empty, may be disposed of in an approved landfill. (refer to 3.2.9) Use of Exterminator Application of the appropriate pesticide, along with all handling and disposal of any associated wastes, by a contracted extermination service is the preferred method of using pesticides. If a service is used, the contractor shall always be required to handle and dispose of any waste from QatarEnergy-IS facilities without QatarEnergy-IS holding responsibility for that waste. When possible, purchase only the pesticide that is used, leaving the contractor the owner of unused pesticide. If pesticides are used directly by QatarEnergy-IS personnel, follow the manufacturer's instructions for storage and application. Spill Requirements If spilled, immediately contact the HSE Department to determine whether the spill needs to be reported to an agency. Contact HSE Department or check MSDS for appropriate personal protective equipment (PPE). Contact HSE Department for guidance on disposal of unusable pesticide and contaminated soil.
Preferred PPE	See MSDS for PPE requirements of particular pesticide, herbicide or biocide. Minimum Safety Glasses and rubber gloves.
Preferred Disposal Method	Removal and disposal by pest control contractor. Where this is not practicable, before disposal of any pesticide waste, consult the HSE Department for instructions.

23	Pyrotechnics - Flares, Smoke Canisters, etc.
Description	Includes old flares (parachute flares and hand flares), smoke canisters and other emergency pyrotechnics which are used offshore and which have become out-of-date. These typically come from emergency stores offshore
Sources	Out of date stock.
Category	Hazardous or Non-Hazardous
Handling and Disposal	Historically, pyrotechnic devices were used in drills and training exercises rather as the expiration date drew near. This is not now possible. Disposal of such devices in Qatar is usually affected with co-operation from the military authorities. Flares or smoke canisters which cannot be used shall be shipped in a specially marked and manifested container. The container shall be of robust, metal construction and be marked with "explosives or flammable - no smoking". Transport of other types of pyrotechnics may require special magazines or containers and shall be considered on a case-by-case basis. Shipment of these materials shall be the responsibility of specialist contractors.
Preferred PPE	Eyes: Goggles when handling broken containers. Normal good room ventilation Gloves: None required unless prolonged handling is anticipated. Respirator if contents become airborne.
Preferred Disposal	Disposal by Ministry of Interior.

24	Refuse
Description	Refuse is the general term for basic trash and inert solid waste and is not usually specific to oil field operations Examples of refuse include garbage, debris, office trash, packing material, paper, lumber scraps, shop rubbish, vegetation, and kitchen waste (including food).
Sources	Canteens, offices, living quarters
Category	Non-hazardous
Handling and Disposal	Be sure all free liquids from a container are used or drained before throwing an empty can or container into the trash receptacle. Use trash skips or, preferably, a trash compactor. Where recycling is

24	Refuse
	available, segregate paper, aluminium and any other recyclable material Food waste shall not be included in this category. Food waste shall be ground up and discharged overboard provided it meets the ROPME requirement (i.e., less than 25 mm in size). Food waste is only the food itself and not any paper or plastic. Food items that cannot be ground (e.g., bones), shall be disposed of in trash. Non-food waste from the galley shall be compacted and sent to shore in skips. Non-food waste shall not be thrown overboard. Placing in trash skips is preferred. To ensure that waste is not blown overboard it shall be collected in plastic bags and placed in the trash skips. All storage containers shall be leak-proof and shall not be unsightly. All storage containers shall be tightly covered fly-proof and rodent-proof. Consideration shall be given to the segregation at source of prime recyclables including metals, plastics, glass, paper and cardboard. If segregated these could be placed in clear or colour coded bags and then into a separate skip for transfer to shore for recycling
Preferred PPE	Cotton Gloves
Preferred Disposal Method	Basel Waste Management Code: R4 & R5 - Recyclables to approved recycling contractors. D1 (for non-recyclable wastes) - Disposal at ME approved municipal landfills.

25	Rig wash
Description	Waste generated from cleaning materials made of detergent and water to aid in the cleaning of rig floors and associated equipment.
Sources	Washing rig floors and operational areas.
Category	Non-hazardous waste (unless heavy oil/chemical contamination).
Handling and Disposal	Contain within the rig containment system and place in storage tank if rig is so equipped. Some rigs tie directly into the open drain system of the platform, in which case the fluid can be flushed into this system.
Preferred PPE	As per the MSDS for the Rig Wash product.

Preferred Disposal Method	Place in produced water stream (See 3.2.20). Collect and dispose with drilling fluids. If this is not practicable, place in drain system.
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26	Sewage
Description	Refers to waste from lavatories, toilets, and showers.
Sources	Sanitation services.
Category	Non-hazardous waste
Handling and Disposal	Treat through a waste unit approved QatarEnergy be macerated before discharge overboard.
Preferred PPE	Rubber gloves. Safety Glasses
Preferred Disposal Method	Basel Waste Management Code: D9 I D7 Marine discharge after on-board sewage pre-treatment.

27	Slop Oil
Description	Slop oil generally refers to oil recovered or separated from the production stream or a waste oil from a part of the separation process. It may also refer to any oil/water mixture or to contaminated oil.
Sources	Production operations, maintenance
Category	Special and/or hazardous depending on the contents
Handling and Disposal	Recover as much as possible through separation to be recycled back to the production stream. Care shall be taken to collect all slop oil so that it does not leak into the sea.
Preferred PPE	Safety Glasses, Neoprene Gloves and Apron, Rubber Boots if a large quantity will be handled
Preferred Disposal Method	Basel Waste Management Code: R3 (preferred), D10 Any waste oil drums that come to Halul are emptied into the QatarEnergy-API Pit and then pumped onto either the PS1, PS2 or PS3 MOLs going to their respective QatarEnergy receiving tanks. Whatever cannot be re-injected into the MOL shall be drummed and sent to Halul Island. See QatarEnergy Advisory on crude oil disposal

28	Solvents
Description	Solvents are used for a variety of purposes, usually associated with cleaning or maintenance tasks. Examples: Paint thinner, WD-40, naphthalene, oil- based cleaning fluid, degreaser.
Sources	Equipment cleaning / maintenance.
Category	Considered hazardous waste if: Contains 10% or more of trichlorethylene, toluene, benzene, methyl ethyl ketone (MEK), 1.1.1. trichloroethane, or methylene chloride Contains 100% xylene, acetone, or methanol; or Fails the EPA ignitability test /flash point less than 60°C /140° F\.
Handling and Disposal	Check MSDS and container label to determine the container contents. Consult with HSE or Corrosion Engineer to determine if the solvent is hazardous after use. Use water-based solvents or soap cleaners as substitutes for hazardous solvents where possible. (High pressure water washers are also a substitute for using solvents.) Use all the solvent in a container, ensuring no free residue remains. Minimize the amount of solvent being lost during cleaning or maintenance. Where practical, use drip pans or other means of catching the solvent. Do not discharge spent solvents into an open drain system. Store spent solvents in correctly labelled solvent containers.
Preferred PPE	Respiratory: NIOSH-approved respirator shall be worn if OSHA threshold limits exceeded. Eye Protection: goggles with side shield. Glove: Solvent- resistant rubber or neoprene Other Clothing and Equipment: Safety eye bath nearby. Ventilation: Local and mechanical exhaust is required.
Preferred Disposal Method	Basel Waste Management Code: R2 (preferred), D10 Add spent solvents to an oil Production tank; or recycle solvents which were

29	Sorbent Pads
Description	Sorbent materials (pads or tubes) that are contaminated by the clean-up of oil spills and other leaks on a rig or platform
Sources	Oil spill, chemical clean-up
Category	Normally hazardous wastes; may be non-hazardous waste after "weathering".
Handling and Disposal	Where possible wring (squeeze) out the hydrocarbons and place the liquid in the hydrocarbon disposal tank (used oil, production oil tank).
Preferred PPE	Depends on type of oil, chemical being absorbed. Refer to MSDS of spilled liquid
Preferred Disposal Method	Basel Waste Management Code: D10 / D1 (If minimal mobile oil content) Drum in sealed drums, label as to contents, and send to shore base. These materials shall preferably be disposed of by high temperature incineration. Where this is not practicable, e.g., due to lack of incineration capacity, wring out the hydrocarbons and return the hydrocarbons to the production system or place in the open drain system. Allow sorbent materials to "weather" and drain and then place in the industrial trash when they are no longer flammable. "Weathering" is the process of leaving the sorbent materials exposed to the elements to allow natural processes to degrade any hydrocarbons until only non-mobile hydrocarbon constituents remain

30	Spent Caustic and Caustic Spills
Description	Caustic waste shall always be neutralized prior to disposal. Caustic contaminated soil shall not be removed until its pH tests less than 11.5. The natural buffering capacity of soil may neutralize small spills
Sources	Drilling or cleaning fluids
Category	Hazardous waste if pH $\geq$ 11.5 (according to Basel Convention).
Handling and Disposal	Pick up as much spilled caustic as possible. Neutralize to pH between 8 and 10 as soon as possible with a weak acid.
Preferred PPE	Goggles, Face Shield, Chemical Apron, Neoprene Gloves, Neoprene Boots and other as provided in product MSDS

Preferred Disposal Method	Basel Waste Management Code: D9 + D7 After neutralization dispose in an injection or disposal well or discharge overboard. Discharge overboard is only permitted if there are no other contaminants present.
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31	Spilled Glycol
Description	Glycol, which can be referred to as: EG [ethylene glycol], DEG [diethylene glycol] or TEG [tri ethylene glycol] is a treating chemical used for dehydration of natural gas. The glycol is regenerated and additional glycol is added to the system. Occasionally all the glycol is replaced in a system. Glycol does biodegrade in the environment but is toxic to marine life in high concentrations.
Sources	Spillage Clean-up.
Category	Non-hazardous in produced water.
Handling and Disposal	Eliminate spills. Recycle glycol by filtration and regeneration.
Preferred PPE	PVC, Latex, Nitrile or Neoprene Gloves, Goggles or Face Shield, Particle Mask
Preferred Disposal Method	Basel Waste Management Code: R3 (preferred) I D7 Send used glycol, via the sump and MOL, to Halul Island. Unusable liquid may be placed in a disposal well.

32	Tank Bottoms and Basic Sediment {BS}
Description	Tank bottoms and Basic Sediment (BS) are oil field terms referring to solid and emulsified contaminants that settle out of produced oil into tanks, pits and process vessels. It is normally a viscous liquid heavily laden with solids and often entrained with water.
Sources	Tank bottoms
Category	Non-hazardous waste unless containing NORM, Iron Sulphide or mobile hydrocarbons.
Handling and Disposal	Removal from the vessel shall be done in a manner where no spillage occurs. Use of drip pans or catchment vessels is recommended. If NORM or iron sulphide is present, treat as the respective waste.
Preferred PPE	For Tank Cleaning, full body Tyvek Suit, SCBA or Airline Respirator, Neoprene Gloves, Rubber Boots,

Preferred Disposal Method	Basel Waste Management Code: D10 high temperature incineration (with gas cleaning)/ D1 - Landfill (only if insignificant oil content). Ship in drums to Halul Island or onshore (drums shall be labelled with the type of waste) for disposal.
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33	Unused Acidizing Fluid
Description	Acidizing fluid remaining after the stimulation of a well can be a hazardous waste.
Sources	Workovers
Category	The chemical is considered hazardous if it exhibits one of the hazardous characteristics listed in the hazardous waste definition (e.g. Corrosiveness - pH less than or equal to 2).
Handling and Disposal	Use the extra fluid in the treatment of the well, even if in excess of the design volume. Verify with Production/Facilities Engineering that this is an acceptable option.
Preferred PPE	Chemical goggles, Face Shield, Chemical Apron, Neoprene Gloves and other as provided in the product MSDS
Preferred Disposal Method	The service company contracted to undertake the acid or frac. work shall be required to remove unused solution. If this is not practicable, then waste acid can be neutralized and disposed of as a non-hazardous waste (see 3.2.2).

34	Unused or Unusable Chemical
Description	Unused chemicals, or chemicals which are unusable (e.g., due to shelf-life expiry) are considered wastes. Examples are: Full or partially full containers of unused chemical; large volume chemicals in drums, bulk storage, sacks, etc.; small volume chemicals used in labs and maintenance shops
Sources	Supply, warehousing, maintenance.
Category	The chemical is considered hazardous if it contains hazardous substances listed by EPA's RCRA (define EPA and RCRA in definitions) regulation or exhibits one of the hazardous characteristics listed in the RCRA regulations; or if the chemical would be considered a hazardous waste in accordance with the Basel Convention. (See the def. of hazardous waste in section 3.0)

	Chemicals which no longer have a use at a specific facility shall not be disposed of until all options discussed below are exhausted.
Handling and Disposal	If a chemical will not be used in the foreseeable future at a location, the following action is recommended: Return to vendor, if they will accept it. Find alternative use for it at that facility or at another QatarEnergy-ISfacility. Consider well treatment outside the normal treatment schedule, as long as no damage to the well is incurred. Disposal is determined by the waste category of the chemical. Consult with the HSE department about approved disposal options.
Preferred PPE	As per the associated MSDS sheet.
Preferred Disposal Method	Treat as appropriate for chemical in question.

35	Used Filters
Description	Filters are any material used in a process unit or engine which remove solid contaminants from a liquid or gas stream. (Examples: Glycol filters, produced water filters, lube/fuel oil filters on engines or compressors, vehicle filters.)
Sources	Vehicle and plant/equipment maintenance.
Category	Non-hazardous/hazardous depending upon entrained contaminant.
Handling and Disposal	After removing spent filters from a unit, drain all free liquids from the filter. Catch any drained fluids in a container and recycle them. Filters can be sent to a municipal landfill approved by MOECC.
Preferred PPE	Depending on entrained contaminant use Safety Glasses or Goggles, Face-shield if a splash hazard, Neoprene Gloves, Chemical Apron, and rubber boots depending upon quantity to be handled and other as provided in the product MSDS
Preferred Disposal Method	Basel Waste Management Code: D10 Oil filters shall be disposed of by high temperature incineration where practicable, with prior consent from MOECC

36	Waste Paint
Description	Unwanted paint other than the film left in an otherwise empty can. Paint thinners shall be treated as solvents as in Section 3.2.27
Sources	Fabrication I maintenance.
Category	May be hazardous or non-hazardous
Handling and Disposal	All paint shall be used up whenever possible. All paint shall be removed from the bucket.
Preferred PPE	If waste paint is dry then normal PPE of Safety Glasses and Cotton Gloves. Otherwise see "Solvents"
Preferred Disposal Method	Basel Waste Management Code: D1 Empty paint buckets may be crushed and taken to a landfill. Empty Paint cans / bucket maybe returned to the supplier for recycling

37	Used Lube/ Hydraulic Oil
Description	Oil that has been utilized in production or other equipment for lubrication or other purposes and has been drained or has leaked from the equipment. Used oil includes used lube oil and hydraulic fluid.
Sources	Maintenance, for example of gearboxes, compressors, hydraulic systems.
Category	Hazardous waste
Handling and Disposal	Store in a metal or plastic drum with bungs tightly in place or a tank of good integrity. Container shall be labelled with the words "Used Oil" and also with the source e.g., "Generator gear box"
Preferred PPE	Safety Glasses, Neoprene Gloves and Apron, Rubber Boots if a large quantity will be handled
Preferred Disposal Method	Basel Waste Management Code: R3 (preferred)/ D10 Any waste oil drums that come to Halul are emptied into the QatarEnergy-API Pit and then pumped onto either the PS1, PS2 or PS3 MOLs going to their respective QatarEnergy-receiving tanks. Oil recovery, recycling by MOECC approved contractor. High temperature incineration, with prior consent from MOECC.

38	Waste Pipe Dope
Description	Unwanted dope normally used on the threads of drill pipe when making up. May contain lead.
Sources	Pipe assembly.
Category	Normally non-hazardous but may be hazardous if contains lead.
Handling and Disposal	Unleaded dope shall be used. All dope shall be removed from the bucket with a spatula or scraper.
Preferred PPE	Coveralls, Safety Glasses, Neoprene Gloves and Apron if handling large quantity.
Preferred Disposal Method	Empty pipe dope buckets may be crushed and taken to a landfill. If disposal of unused pipe dope is necessary call the HSE Department.

39	Workover Fluids
Description	Primarily produced fluids with additives for special purposes such as weighting agents, surfactants, muds, acids, inhibitors, gels, solvents, and other materials.
Sources	Workovers
Category	Generally non-hazardous waste depending on additives.
Handling and Disposal	Treat as unused or unusable chemical.
Preferred PPE	Follow PPE requirements of associated MSDSs of additives used.
Preferred Disposal Method	Return to the crude oil production system via MOL or by shipping in drums to Halul Island.

40	Empty Plastic Containers/Bottles
Description	Empty plastic containers/bottles which previously contained chemicals or material used at a facility shall only be considered empty if all the contents have been removed. They shall be disposed of in accordance with Section 3.9 (Empty Drums). However, empty containers/bottles previously used for potable water, and other plastic containers not containing hazardous materials shall also be disposed of carefully.

40	Empty Plastic Containers/Bottles
	The bottles are not inherently hazardous but there is a risk of their being used for incorrect purposes i.e., to store drinking water, chemicals for which they were not designed, and so on.
Sources	Chemical usage
Category	Non-hazardous waste unless they contain residues of hazardous substances.
Handling and Disposal	The bottles shall be crushed to reduce volume. Almost all potable water bottles are now designed to be crushed to a small fraction of their full volume. Ship to shore as non-hazardous waste. Do NOT incinerate unless a purpose-built high temperature incinerator is available. Low temperature incineration may lead to the release of dioxins.
Preferred PPE	No PPE for empty water bottles Neoprene gloves and safety glasses for empty plastic chemical containers
Preferred Disposal Method	Basel Waste Management Code: R3 (preferred) if not practicable D10 / D1 Recycle if facilities available. High temperature incineration. If neither of the above options are available then wastes may be land filled.

41	Empty Aerosol Cans
Description	Aerosol cans may, originally, contain hazardous or non-hazardous materials. The propellant may or may not be flammable. However, the can, even when empty presents a potential hazard due to the residual internal pressure
Sources	Maintenance.
Category	Generally non-hazardous waste unless they contained hazardous substances or utilized a hazardous (e.g., flammable) propellant or a CFC propellant.
Handling and Disposal	Segregate and ship ashore as hazardous waste
Preferred PPE	Cotton Gloves, Safety Glasses

Preferred Disposal Method	Basel Waste Management Code: D10 / D1 If practicable, puncture in dedicated aerosol destructor and incinerate residuals, otherwise Landfill
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WASTE MANAGEMENT PROCEDURE

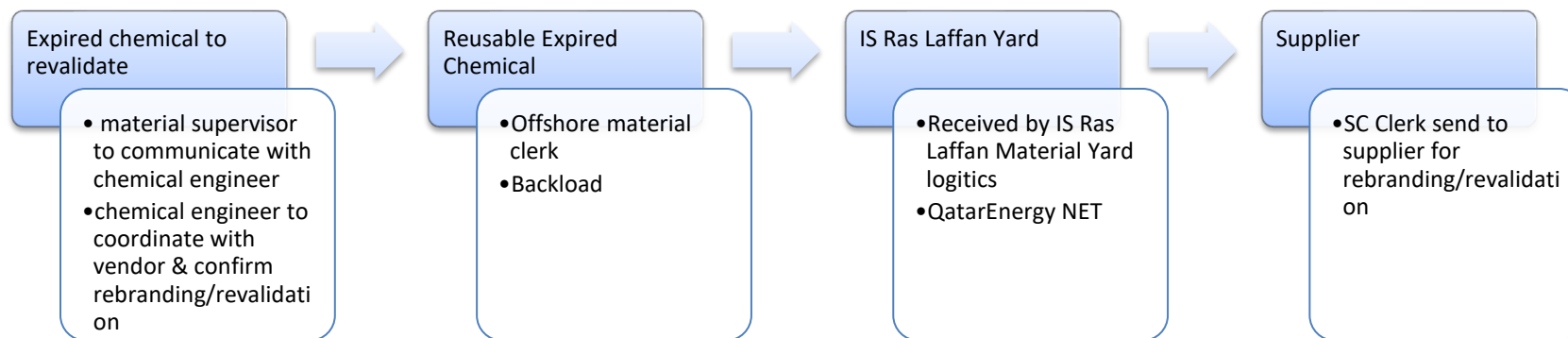
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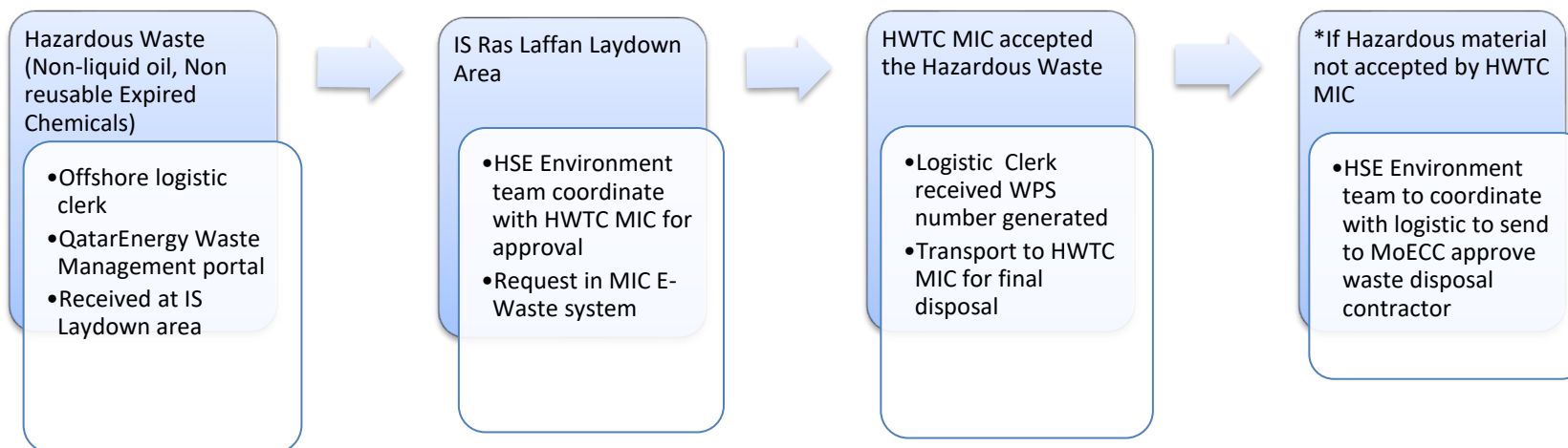
APPENDIX C – IS WASTE MANAGEMENT WORKFLOW

Appendix C: Idd El Shargi Waste Workflow.

1. Revalidation of Expired Chemicals Workflow.



2. Hazardous Waste (Non-liquid oil, Non reusable Expired Chemicals) Workflow



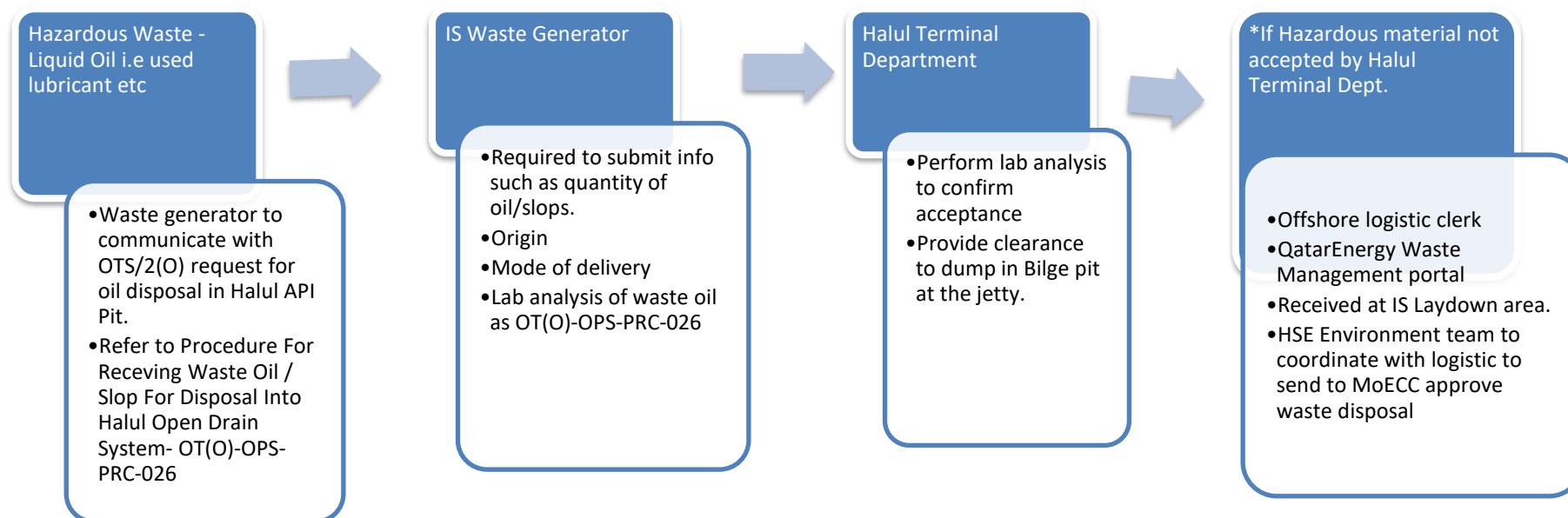


WASTE MANAGEMENT PROCEDURE

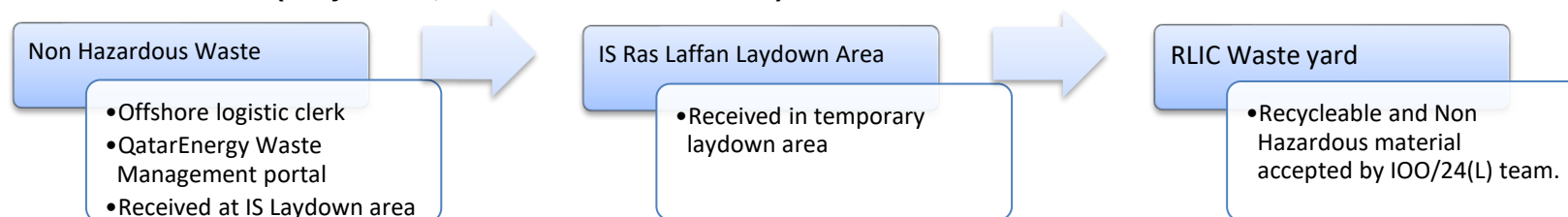
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3. Hazardous Waste (Liquid oil) Workflow



4. Non Hazardous Waste (Recycleable, Domestic and Other Waste) Workflow





APPENDIX D - SKIP LABELS

INERT NON-HAZARDOUS WASTE SKIP													
<u>ONLY</u> THE FOLLOWING MATERIALS ARE TO BE PLACED IN THIS SKIP ***<u>NO</u> CHEMICALS and <u>NO</u> LIQUIDS*** <table border="0"> <tr> <td>▪ WOOD</td> <td>▪ RUBBER</td> </tr> <tr> <td>▪ DUST FILTERS</td> <td>▪ GENERAL GARBAGE</td> </tr> <tr> <td>▪ CABLE INSULATION</td> <td>▪ COTTON WASTE</td> </tr> <tr> <td>▪ FOOD WASTE</td> <td>▪ BUILDING DEBRIS</td> </tr> <tr> <td>▪ EMPTY BOTTLES, CUPS, FOOD WRAPPERS, PLASTIC SPOONS ETC.</td> <td>▪ FURNITURE</td> </tr> <tr> <td></td> <td>▪ OFFICE WASTE, PAPER, CARDBOARD</td> </tr> </table>		▪ WOOD	▪ RUBBER	▪ DUST FILTERS	▪ GENERAL GARBAGE	▪ CABLE INSULATION	▪ COTTON WASTE	▪ FOOD WASTE	▪ BUILDING DEBRIS	▪ EMPTY BOTTLES, CUPS, FOOD WRAPPERS, PLASTIC SPOONS ETC.	▪ FURNITURE		▪ OFFICE WASTE, PAPER, CARDBOARD
▪ WOOD	▪ RUBBER												
▪ DUST FILTERS	▪ GENERAL GARBAGE												
▪ CABLE INSULATION	▪ COTTON WASTE												
▪ FOOD WASTE	▪ BUILDING DEBRIS												
▪ EMPTY BOTTLES, CUPS, FOOD WRAPPERS, PLASTIC SPOONS ETC.	▪ FURNITURE												
	▪ OFFICE WASTE, PAPER, CARDBOARD												

RECYCLABLE METALS SKIP											
<u>ONLY</u> THE FOLLOWING MATERIALS ARE TO BE PLACED IN THIS SKIP ***<u>NO</u> CHEMICALS and <u>NO</u> LIQUIDS*** <table border="0"> <tr> <td>▪ ELECTRICAL PARTS (RELAYS, CONTACTORS, MCBs, FUSES, FITTINGS ETC.)</td> <td>▪ STEEL / COPPER / BRASS / STAINLESS STEEL / KUNIFER / ALUMINUM</td> </tr> <tr> <td>▪ GRINDING WHEELS</td> <td>▪ EMPTY FOOD CANS</td> </tr> <tr> <td>▪ MOTOR BEARINGS</td> <td>▪ CONSTRUCTION WASTE (VALVES, PIPES, BOLTS, ETC.)</td> </tr> <tr> <td>▪ GAUGES</td> <td>▪ SCRAP ELECTRICAL CABLE</td> </tr> <tr> <td colspan="2">▪ SCRAP SLINGS, SHACKLES AND OTHER LIFTING DEVICES (TO BE DESTROYED PRIOR TO DISPOSAL)</td> </tr> </table>		▪ ELECTRICAL PARTS (RELAYS, CONTACTORS, MCBs, FUSES, FITTINGS ETC.)	▪ STEEL / COPPER / BRASS / STAINLESS STEEL / KUNIFER / ALUMINUM	▪ GRINDING WHEELS	▪ EMPTY FOOD CANS	▪ MOTOR BEARINGS	▪ CONSTRUCTION WASTE (VALVES, PIPES, BOLTS, ETC.)	▪ GAUGES	▪ SCRAP ELECTRICAL CABLE	▪ SCRAP SLINGS, SHACKLES AND OTHER LIFTING DEVICES (TO BE DESTROYED PRIOR TO DISPOSAL)	
▪ ELECTRICAL PARTS (RELAYS, CONTACTORS, MCBs, FUSES, FITTINGS ETC.)	▪ STEEL / COPPER / BRASS / STAINLESS STEEL / KUNIFER / ALUMINUM										
▪ GRINDING WHEELS	▪ EMPTY FOOD CANS										
▪ MOTOR BEARINGS	▪ CONSTRUCTION WASTE (VALVES, PIPES, BOLTS, ETC.)										
▪ GAUGES	▪ SCRAP ELECTRICAL CABLE										
▪ SCRAP SLINGS, SHACKLES AND OTHER LIFTING DEVICES (TO BE DESTROYED PRIOR TO DISPOSAL)											



HAZARDOUS WASTE SKIP

THE FOLLOWING MATERIALS ARE TO BE PLACED IN THIS SKIP ONLY
AFTER CONSULTING WITH THE HES ADVISOR and MATERIALS MAN.

CONTACT ONSHORE HES FOR MORE INFORMATION

- | | |
|---|--|
| ▪ FULL / PARTIALLY FULL
DRUMS OF CHEMICALS | ▪ USED / EMPTY CHEMICAL
CONTAINERS |
| ▪ HYDROCARBON SLUDGE /
WASTES | ▪ INSULATION MATERIALS
(non-asbestos) |
| ▪ AEROSOL CANS | ▪ EMPTY PAINT CANS |
| ▪ EMPTY GAS CYLINDERS | ▪ USED BATTERIES |
| ▪ FILTER ELEMENTS | ▪ USED OIL GASKETS |

**WASTE MANAGEMENT PROCEDURE****DOC NO: IS-HSE-PRC-012****REV. 02****APPENDIX E - MIC-HWTC WASTE PROFILE SHEET: IS-HSE-PRC-012-E****HAZARDOUS WASTE TREATMENT CENTER****MESAIEED INDUSTRIAL CITY, QATAR****WASTE PROFILE SHEET**

Generator Identification No.: _____ WPS No.: _____

(Generator Identification No. and Waste Profile Sheet (WPS) No. - To be assigned by MIC - Waste Management Dept.)

A. General Information

1. Company Name : _____
2. HSE/ENV. Section Head Name : _____
3. Contact No : _____ 4. Fax No : _____
5. Generator Dept : _____ 6. Unit : _____
7. Equipment No : _____

B. Name and Source of Waste

1. Name of the Waste : _____
2. Describe the Process Generating the Waste (if required use / attach additional sheet) :

C. Physical Characteristics of Waste

1. Colour : 	2. Does the waste have a strong odour ? Yes ☐ No ☐ If Yes, describe : _____	3. Physical State : Solid ☐ Liquid ☐ Sludge ☐ Power ☐	4. Specific Gravity : _____	5. Free Flowing Liquid : Yes ☐ No ☐
6. pH : N/A	7. Density : _____	8. % Solids : _____	9. Melting Point : _____	10. Boiling Point : Not applicable
11. Liquid Flash Point : ☐ < 18 deg C ☐ 18 ~ 23 deg C ☐ 23 ~ 61 deg C ☐ > 61 deg C ☐ None ☐ Open Cup ☐ Close Cup				

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**WASTE MANAGEMENT PROCEDURE****DOC NO: IS-HSE-PRC-012****REV. 02****APPENDIX E - MIC-HWTF Waste Profile Sheet (Cont'd)****Generator Identification No.:** _____**WPS No.:** _____

(Generator Identification No. and Waste Profile Sheet (WPS) No. - To be assigned by MIC - Waste Management Dept.)

D. Chemical Composition	Range		E. Metals: Indicate if this waste contains any of the following		
	Min %	Max %	Metal	Less than	Actual
			Arsenic	< 5 PPM	
			Barium	< 100 PPM	%
%			Cadmium	< 1 PPM	
%			Chromium (Total)	< 5 PPM	
		%	Lead	< 5 PPM	
%	%		Mercury	< 0.2 PPM	
		%	Selenium	< 1 PPM	
%	%		Silver	< 5 PPM	

F. Other Hazardous Characteristics							
Identification of this waste any of the following				Indicate if this waste contains any of the following			
Corrosive	☐	Toxic	☐		None	Less than	Actual
Explosive	☐	Oxidising	☐	PCBs		< 50 PPM	PPM
Flammable	☐	Pyrophoric	☐	Cyanides		< 50 PPM	% ☐ PPM
Ignitable	☐	Reactive	☐	Phenols	% ☐	< 50 PPM	PPM
Infectious	☐	Radioactive	☐	Sulphides		< 50 PPM	PPM
Irritant	☐	Shock Sensitive	☐	Benzene	% ☐		PPM
Other	☐	Pl. specify :		Toluene			PPM
If reactive, pl specify :				Xylene			PPM

G. Incineration Parameters :				H. Other Parameters :	
	None	< Than	Actual		
Bromine		< 1 %	% ☐	1. Heat Value (Kcal/kg)	
Chlorine		< 5 %	% ☐	2. Water Content (%)	% ☐
Fluorine		< 1 %	% ☐	3. Viscosity (cps @ 50 deg C)	
Sulphur		< 3 %	% ☐	4. Ash (%)	
Sodium	%	< 1 %	%	5. Settle able solids (%)	
Potassium		< 1 %	% ☐	6. Is liquid waste pumpable?	Yes ☐ No ☐
Phosphorus		< 1 %	% ☐	7. Can this waste be heated to improve the flow?	Yes ☐ No ☐
				8. Is the waste soluble in water?	Yes ☐ No ☐

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**WASTE MANAGEMENT PROCEDURE****DOC NO: IS-HSE-PRC-012****REV. 02****APPENDIX E - MIC-HWTF Waste Profile Sheet (Cont'd)**

Generator Identification No.: _____ **WPS No.:** _____
 (Generator Identification No. and Waste Profile Sheet (WPS) No. - To be assigned by MIC - Waste Management Dept.)

I. General Information :	
1. Anticipated Quantity:	_____ Normal _____ Max _____
2. Frequency of Generation:	_____
3. Method of collection:	Drum ☐ Bag ☐ Bulk Container ☐ Vacuum Tanker ☐ Other ☐
If others, Specify : _____	

J. Safety Information :	
1. Category:	_____
2. NFPA Classification :	
Health:	_____ Flammability: _____ Reactivity: _____ Special Hazard: _____
3. Describe any other Hazards associated with waste :	_____

K. Special Handling Information : (Please attach a copy of MSDS)	

L. Certification on the information :	
This is to certify that the provided information are accurate and which respect the true forms the waste. Duly filled waste profile sheet should be approved by component authority of the waste generator organization.	

Prepared by
(By the waste generator)

Approved by
(By the waste generator)

Signature :	_____	_____
Name :	_____	_____
Designation :	_____	_____
Date :	_____	_____

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	WASTE MANAGEMENT PROCEDURE DOC NO: IS-HSE-PRC-012 REV. 02
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APPENDIX F – REVISION HISTORY LOG**Revision Number: 02****Date: 28/05/2025**

Item Revised	Revision Description	Page No.
3.0, 4.11 & 4.12.	Responsibility for conducting audits of waste disposal locations has been removed since this is now under PMM department responsibility.	2,10, 11
Remarks:		

APPENDIX G – LIST OF ABBREVIATIONS

Term	Definition
CTO	Consent to Operate
MIC HWTC	Mesaieed Industrial City - Hazardous Waste Treatment Centre
MOECC	Ministry of Environment & Climate Change
MOL	Main Oil Line
MSDS	Material Safety Data Sheet
NORM	Naturally Occurring Radioactive Material(s)
ROPME	Regional Organization for the Protection of the Marine Environment